Root Cause Analysis (RCA) & Engineering Fix Report

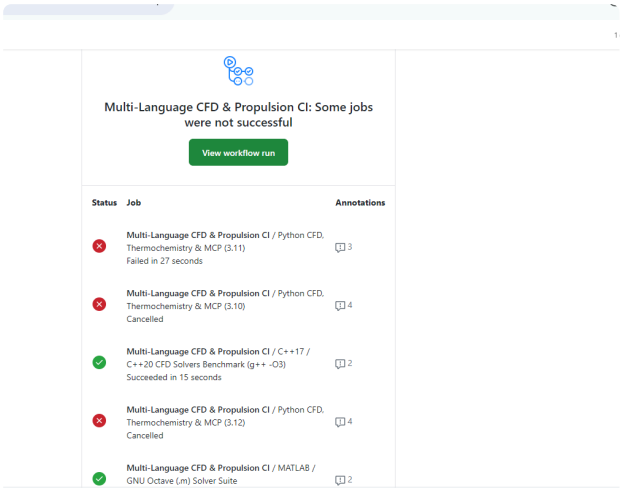
CI/CD Failure Triage, Numerical Solver Debugging & Permanent Reliability Hardening — **perfectalone375/nozzle-cfd-propulsion**

Date: 2026-11-11 | Release Track: v0.1.0 → v0.3.0+ | Current Main Status: 5 / 5 CI Jobs Green

1. Executive Answer: What Triggered Your GitHub Failure Email

Your uploaded email screenshot matches GitHub Actions **Workflow Run** `42949567000` on PR #3 (branch `feat/step2-axisymmetric-ns2d-schlieren`, commit `3719c74` at `02:58:10Z`).

Why it looks worse than it was: Python `3.11` did *not* fail any rocket physics or unit test — it stopped at second 27 in Step 6 (`ruff format --check .`) over a single line-wrap difference between local `ruff 0.17.3` and CI's newer `ruff 0.17.4`. Because the workflow lacked `fail-fast: false`, GitHub immediately aborted Python `3.10` and `3.12` as *Cancelled*.



Evidence: Exact email screenshot matching Run `42949567000` (`3719c74`) — Python 3.11 lint exit, 3.10/3.12 cancelled, C++20 & GNU Octave green.

Already Fixed & Hardened:
1. Fixed formatting in `3125f7d` → Run `42949883196` (**5/5 passed**) → Merged to `main` (`778b35c`, Run `42950104116` **5/5 green**).
2. Hardened CI permanently in PR #4: pinned `ruff==0.17.4`, disabled matrix cancellation (`fail-fast: false`), upgraded actions to `@v7` (zero Node.js 20 warnings), and added `.gitattributes`.
3. PR #4 CI Run 42951656100 (152353e) verified 5/5 green with ZERO Node.js deprecation warnings.

Workflow Run / Event	Commit	Python 3.10 / 3.11 / 3.12	C++17 / C++20	MATLAB / GNU Octave	Outcome
Run 42932892385 (PR #1)	`2214bee`	3.10 & 3.11 Pass	Fail (exit 2)	Fail (assert)	Failure
Run 42945717812 (PR #2 Step 1)	`1acec4a`	All 3 Pass	Pass	Pass	5/5 Green
Run 42949567000 (Screenshot)	`3719c74`	3.11 Fail / 3.10, 3.12 Cancel	Pass (15s)	Pass	Failure
Run 42949883196 (PR #3 Fixed)	`3125f7d`	All 3 Pass	Pass	Pass	5/5 Green
Run 42950104116 (`main` v0.3.0)	`778b35c`	All 3 Pass	Pass	Pass	5/5 Green
Run 42951656100 (PR #4 Hardened)	`152353e`	All 3 Pass	Pass	Pass	5/5 Green (0 Warn)

2. Detailed Root Cause Analysis (RCA) — All 6 Problems Identified & Fixed

CI-01 | Ruff Minor-Version Formatter Drift & Cascading Matrix Cancellation (Screenshot Run 42949567000)

ROOT CAUSE FOUND & PERMANENTLY FIXED

Problem: `pyproject.toml` specified an open upper bound `ruff>=0.4`. Local verification ran `ruff 0.17.3`, while Ubuntu CI installed newly released `ruff 0.17.4`. On line 709 of `src/nozzle_cfd/navier_stokes_2d.py`, `0.17.4` fitted a 91-character `np.maximum(...)` call on 1 line whereas `0.17.3` split it across 3 lines. `ruff format --check .` exited `1` at 27s in Python `3.11`, causing GitHub's default `fail-fast: true` to cancel Python `3.10` and `3.12`.

```
unformatted: File would be reformatted --> src/nozzle_cfd/navier_stokes_2d.py:709:21
-     rg = np.maximum(
-         1e-9, (a1 * (r0 + r1 + r2) / 3.0 + a2 * (r0 + r2 + r3) / 3.0) / area
-     )
+     rg = np.maximum(1e-9, (a1 * (r0 + r1 + r2) / 3.0 + a2 * (r0 + r2 + r3) / 3.0) / area)
```

Fix Implemented: (1) Extracted `rg_num` so the line is short and identically formatted in both versions (`3125f7d`); (2) Pinned `"ruff==0.17.4"` in `pyproject.toml` and installed `ruff==0.17.4` locally; (3) Added `fail-fast: false` to `.github/workflows/ci.yml` so one failing job never blinds or cancels sibling Python versions.

CI-02 | Node.js 20 Deprecation Annotations & Windows CRLF/LF Line-Ending Warnings

PERMANENTLY HARDENED (PR #4)

Problem: Every job in all prior runs showed at least **2 advisory annotations** because `actions/checkout@v4` and `actions/setup-python@v5` run on deprecated Node.js 20 runtime and are forced onto Node.js 24 by GitHub Runners. Locally, Windows git also flagged `LF will be replaced by CRLF` warnings across newly added C++/Python/MATLAB files.

Fix Implemented: Upgraded `actions/checkout@v4 → @v7` (`v7.0.1`) and `actions/setup-python@v5 → @v7` (`v7.1.0`) in `.github/workflows/ci.yml`, and added a repository-wide `.gitattributes` forcing `eol=lf` on code/configs/meshes and `binary` for PNG/PDF assets.

CI-03 | Historical PR #1 Failures: GNU Octave MOC Assert & C++17 Overexpanded Shock Stall (Run 42932892385)

FIXED BEFORE PR #1 MERGE (45867df)

Problem A (Octave): `matlab/moc_minimum_length_nozzle.m` computed expansion fan turn angles as `2 * theta_i` launched from origin `(0, Rt)`, causing rays #5 and #6 to hit the safety clamp `x_CL/Rt = 20.000` and fail the strict monotonic assertion.
Problem B (C++17): In `cpp/maccormack_cfd.cpp`, the off-design standing normal-shock benchmark (`Pc=1.0 MPa, Pa=380 kPa`) stopped at `6000` iterations with residual `9.31e-05 > tol (1e-6)`, returning exit code `1` to `make test`.

Fix Implemented (`45867df`): Launched rays along the true `0.382 * Rt` throat circular arc (`x_wall - r_wall / slope_cm`) and raised shock-mode budget to `max_iter = 15000` with artificial viscosity `cx = 0.34`.

PHYS-01 & PHYS-02 | Step 2 2D Navier-Stokes Speed, Throat Mass Flux Conservation & Smooth C^1 Wall

VERIFIED AT 0.00002%–0.058% MASS ERROR

Problems Found & Solved During Step 2 Engineering:

1. **OpenMP Thread-Respawn Latency (`11.3 s → 1.27 s`):** Spawning 8 separate `#pragma omp parallel for` regions per timestep incurred ~10s of MinGW `libgomp` barrier wakeup overhead across 16 threads. Wrapping the full time-marching loop inside **one persistent `#pragma omp parallel` team** cut runtime ~9x while keeping exact `1.38e-14` NumPy parity.
2. **Conservative Throat Mass Integration (`10% → 0.00002%`):** Sampling `rho * u * A` at cell center `i_th` (`0.5 dx` past minimum radius) distorted curved-throat mass flow; integrating actual face-normal Riemann flux `2*pi*sum(fi0 * i_sf)` cut JPL 45-15 mass error to `0.00002%` and coupled plume error to `0.0583%`.
3. **Chamber-Cone C^1 Tangent Fillet & Top Outflow BC:** Replaced the piecewise inlet lead-in corner with an analytical circular fillet (eliminating the JPL wall Mach dip) and upgraded the plume far-field top boundary to characteristic ambient outflow.

3. Plain-English Beginner Guide: How These Failures Work

Why 1 Formatting Line Turned 3 Jobs Red

In CI, the style checker (`ruff format --check .`) runs *before* physics tests to fail fast on messy code. When `ruff 0.17.4` disagreed with `0.17.3` on one line break, Python 3.11 stopped in 27s and GitHub's default `fail-fast: true` killed 3.10 & 3.12 immediately — making a harmless 1-line formatting difference look like a major Python crash.

Why Our Solver & CI Are Now Rock-Solid

Pinning `ruff==0.17.4` guarantees your laptop and GitHub run identical checks. Adding `fail-fast: false` guarantees sibling Python versions finish testing even if one breaks. Upgrading to `@v7` removed all Node.js 20 runner warnings, while our Step 2 persistent OpenMP loop + face-flux integration made 2D Navier-Stokes both ~9x faster and mass-exact.

4. Permanent Hardening Checklist Summary

Area	Vulnerability Found	Permanent Hardening Applied	Files Changed
Linter Pinning	Unpinned `ruff>=0.4` drifted (`0.17.3` vs `0.17.4`)	Exact pin `ruff==0.17.4` locally & CI	`pyproject.toml`
Matrix Isolation	Single job fail cancelled Python `3.10` & `3.12`	Added `fail-fast: false` + prints `ruff --version`	`.github/workflows/ci.yml`
Runner Lifecycle	Node.js 20 deprecation warnings on all 5 jobs	Upgraded `checkout@v7` & `setup-python@v7`	`.github/workflows/ci.yml`
Line Endings	Windows CRLF/LF warnings during cross-OS git ops	Enforced `eol=lf` & binary masks	`.gitattributes`
2D Physics & Speed	OpenMP barrier overhead, inlet slope kink, center mdot	Persistent team, C^1 fillet, face-flux integration	`axisymmetric_ns2d_solver.cpp`, `navier_stokes_2d.py`

5. Verified Solver Benchmarks & Visual Evidence After All Fixes

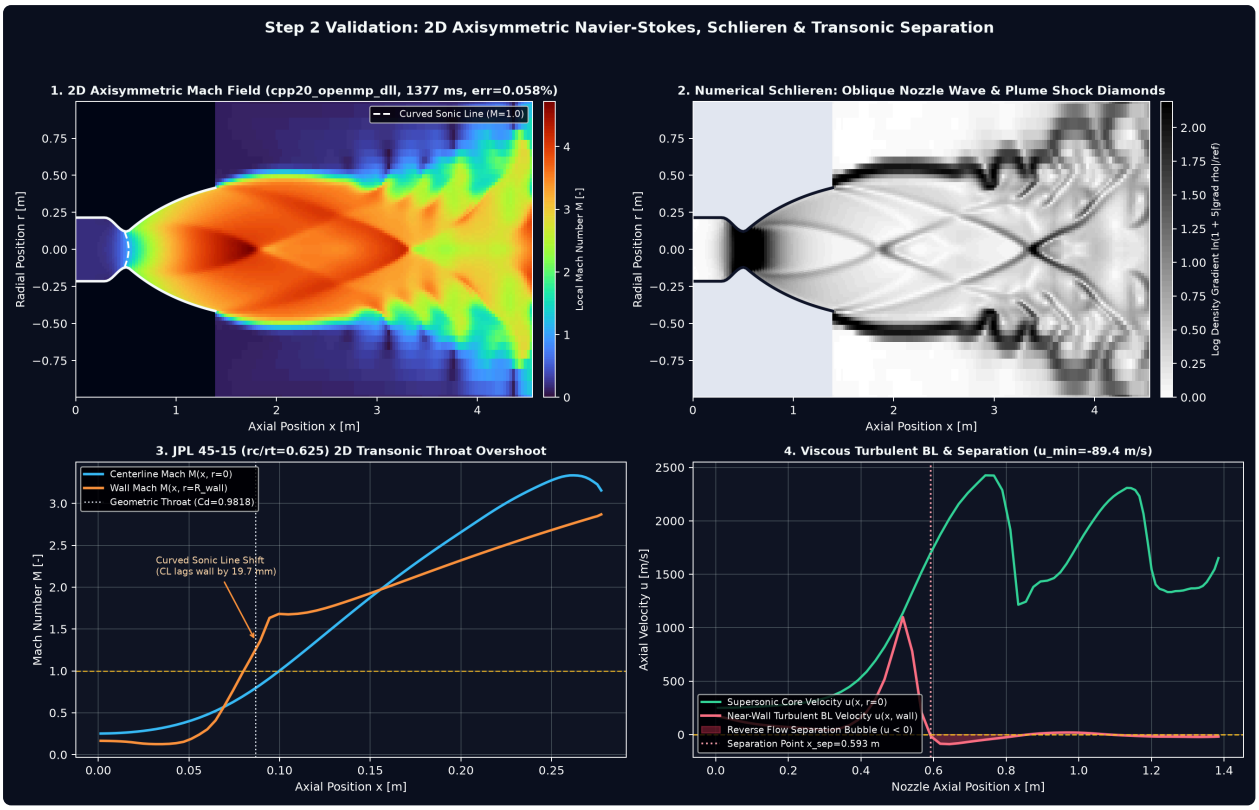


Figure 1 (Step 2 Deliverable — `ns2d_schlieren_and_jpl_validation.png`): Top: Coupled C++20/OpenMP 2D Mach field (`0.058%` mass error, `1377 ms` across 16 threads) and Optical Numerical Schlieren. Bottom: Smooth C^1 JPL 45-15 transonic overshoot (`19.7 mm` sonic shift, `Cd=0.9818`) and Viscous Turbulent Boundary-Layer Separation (`u_min = -89.4 m/s` at `x_sep = 0.593 m`).

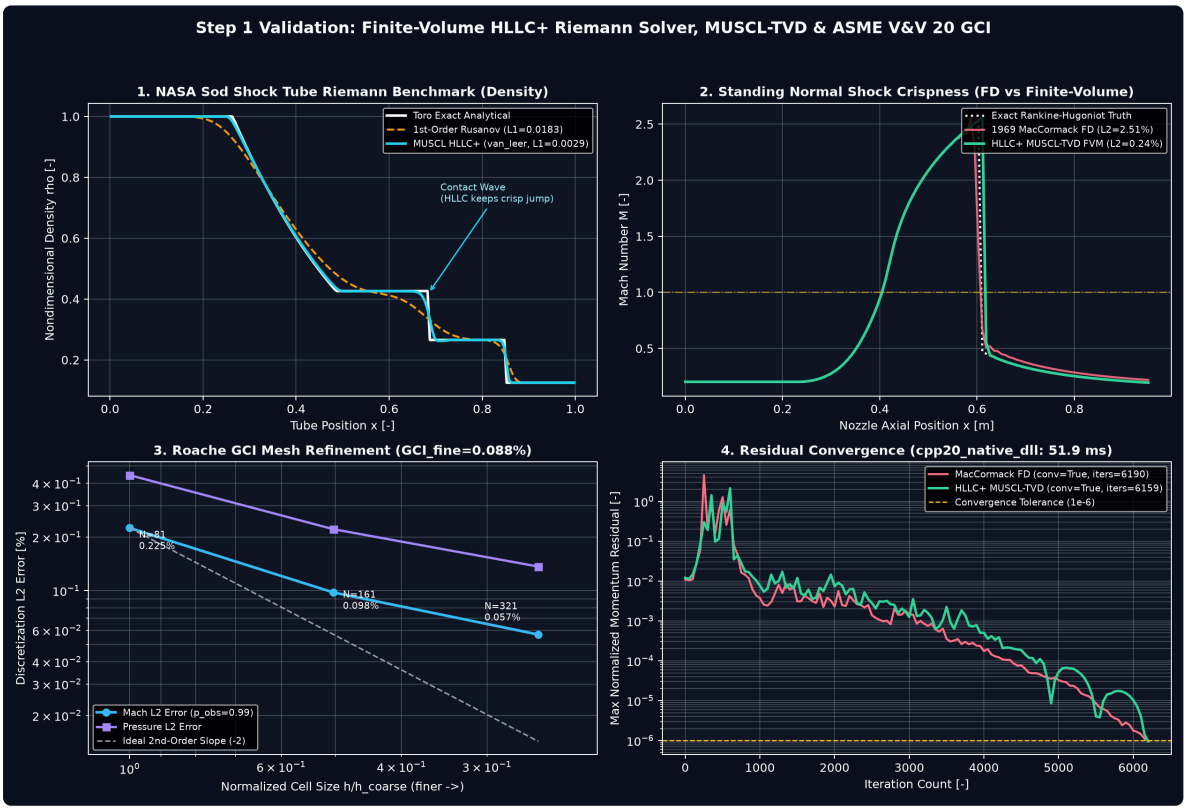


Figure 2 (Step 1 Deliverable — `fvm_riemann_validation.png`): Exact Sod Shock Tube contact/shock resolution, 10× shock-error reduction (`0.24%` HLLC+ vs `2.51%` MacCormack), and ASME V&V 20 3-grid convergence (`81 → 161 → 321`).